

Pass-Through, Welfare, and Incidence under Product Returns

Draft notes

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Abstract

With product returns, local welfare and incidence weight kept units rather than gross sales, and pass-through is unchanged once list price is replaced by effective marginal revenue.

1 Setting

A homogeneous product is sold at list price p . Consumers indexed by i differ in return hassle κ_i , the disutility from returning the product, and in match distribution G_i over post-purchase match quality u . After purchase, type i draws $u \sim G_i$ and keeps the item whenever $u - p \geq -\kappa_i$, earning surplus $u - p$, and returns it otherwise, earning $-\kappa_i$.

Buying at price p yields expected surplus

$$s_i(p) = \mathbb{E}_{u \sim G_i} [\max(u - p, -\kappa_i)].$$

For purchase, all consumer-side heterogeneity summarizes into willingness to pay v_i , defined by $s_i(v_i) = 0$, so type i buys if and only if $p \leq v_i$. Across types, $F(v) = \Pr(v_i \leq v)$, gross demand is $D(p) = 1 - F(p)$, and share $r(p)$ of gross units sold at p are returned.

The product market is perfectly competitive, producer j takes p as given, chooses output at cost $C_j(q_j)$, and pays handling cost $h > 0$ on each returned unit.

2 Consumer side

Write $s(v, p)$ for expected surplus at list price p among types with willingness to pay v , so $s(v, v) = 0$. Consumer surplus integrates surplus over all types who buy,

$$CS(p) = \int_p^\infty s(v, p) dF(v). \tag{1}$$

Differentiating in p gives an extensive-margin term from types entering or leaving the purchase margin and an intensive-margin term from lower surplus among inframarginal buyers,

$$CS'(p) = -s(p, p) f(p) + \int_p^\infty s_p(v, p) dF(v),$$

where $s_p(v, p) \equiv \partial s(v, p) / \partial p$ and f is the density of v_i when F is differentiable. On the extensive margin, $s(p, p) = 0$, so entry and exit contribute nothing at first order. On the intensive margin, $s_p(v, p)$ equals minus the keep probability, because a higher price harms the consumer only when she keeps the purchase,

$$s_p(v, p) = -\Pr(u \geq p - \kappa_i).$$

Aggregating over buyers who purchase then gives

$$CS'(p) = \int_p^\infty s_p(v, p) dF(v) = -D(p)(1 - r(p)),$$

by the definitions of $D(p)$ and $r(p)$. A small price change therefore satisfies

$$dCS = -D(p)(1 - r(p)) dp. \quad (2)$$

The extensive- and intensive-margin decomposition makes the local formula transparent. Consumer surplus changes depend only on the aggregates $D(p)$ and $r(p)$, both at each price and integrated along any path,

$$\Delta CS = - \int_{p_0}^{p_1} D(p)(1 - r(p)) dp.$$

Micro heterogeneity in price responses can be rich. Price p enters each $s_i(p)$ nonlinearly and can shift participation, return decisions, and inframarginal surplus differently across types. Yet once (D, r) are given, this microstructure is irrelevant for dCS and ΔCS . Two economies with the same paths of $D(p)$ and $r(p)$ deliver the same consumer surplus loss, whether high- v_i types adjust return behavior more steeply than low- v_i types or not. Only kept units enter at first order, through $1 - r(p)$, not gross purchases $D(p)$ alone.

Let $\bar{p}$ denote the highest willingness to pay in the population, so $D(\bar{p}) = 0$ and $CS(\bar{p}) = 0$. Integrating the local formula then rewrites consumer surplus as area under kept volume,

$$CS(p) = \int_p^{\bar{p}} D(x)(1 - r(x)) dx. \quad (3)$$

3 Supply side

Returns make list price an imperfect summary of producer payoffs. A gross sale brings revenue p if the consumer keeps the item and costs refund p plus handling h if the item is returned. Among gross sales at list price p , share $r(p)$ are returned, so expected revenue per gross unit—the *effective marginal revenue*—is

$$\mu(p, h) \equiv p - r(p)(p + h). \quad (4)$$

Under perfect competition, firm j takes p as given and therefore faces a fixed number μ in its supply decision. At equilibrium, $\mu = \mu(p, h)$.

Given μ , firm j chooses output to solve

$$\max_{q_j \geq 0} \mu q_j - C_j(q_j), \quad \text{FOC: } C'_j(q_j) = \mu \text{ (if } q_j > 0),$$

with $C_j(0) = 0$. Write $q_j(\mu)$ for firm j 's supply, set to zero when μ is below its minimum marginal cost.

Summing across firms defines market supply and industry profit,

$$S(\mu) \equiv \sum_j q_j(\mu), \quad PS(\mu) \equiv \sum_j [\mu q_j(\mu) - C_j(q_j(\mu))]. \quad (5)$$

Which firms are active at each μ determines both the market quantity and total profit.

Differentiating $PS(\mu)$ with respect to μ gives, for each firm,

$$\frac{d}{d\mu} [\mu q_j - C_j(q_j)] = q_j + q'_j(\mu) [\mu - C'_j(q_j)].$$

Active firms satisfy $C'_j(q_j) = \mu$ at their chosen output, so the adjustment term $q'_j(\mu) [\mu - C'_j(q_j)]$ is zero for each firm—equivalently, the envelope theorem applied to each firm's profit problem gives $d\pi_j/d\mu = q_j(\mu)$. Summing over firms,

$$PS'(\mu) = \sum_j q_j(\mu) = S(\mu).$$

With no production at $\mu = 0$, it follows that industry profit equals the area to the left of the market supply curve,

$$PS(\mu) = \int_0^\mu S(x) dx. \quad (6)$$

At equilibrium $\mu = \mu(p, h)$, $PS = \int_0^{\mu(p, h)} S(x) dx$.

4 Pass-through

Equilibrium satisfies $D(p) = S(\mu(p, h))$. Write $Q \equiv D(p)$. We ask how list price moves when costs rise locally. A uniform one-dollar parallel shift in every firm's marginal production cost shifts market supply at a given effective marginal revenue μ . A one-dollar increase in handling cost h lowers $\mu(p, h)$ at a fixed list price p .

For either shock, write the clearing residual and equilibrium condition

$$m(p, x) \equiv D(p) - S(\mu(p, x), x) = 0, \quad x \in \{c, h\}. \quad (7)$$

Differentiating while keeping $m = 0$ defines pass-through $\rho_x \equiv dp/dx$ by

$$\rho_x = -\frac{m_x}{m_p}, \quad (8)$$

where $m_p \equiv \partial m/\partial p$ and $m_x \equiv \partial m/\partial x$. Since $m = D - S$,

$$m_p = D'(p) - S'(\mu) \mu_p, \quad m_x = -S'(\mu) \mu_x - S_x, \quad (9)$$

with $\mu_p \equiv d\mu/dp$, $\mu_x \equiv \partial\mu/\partial x$ at fixed p , $S'(\mu) \equiv dS/d\mu$, and $S_x \equiv \partial S/\partial x$ at fixed μ . Hence

$$\rho_x = \frac{S'(\mu) \mu_x + S_x}{D'(p) - S'(\mu) \mu_p}. \quad (10)$$

A one-dollar parallel rise in production cost at fixed μ is equivalent to lowering the μ that clears marginal cost by one dollar, so $S_c = -S'(\mu)$. At fixed list price p , $\mu_c = 0$ and $S_h = 0$. Hence

$$\mu_c = 0, \quad S_c = -S'(\mu), \quad \mu_h = -r(p), \quad S_h = 0. \quad (11)$$

Write $\rho_c \equiv dp/dc$ and $\rho_h \equiv dp/dh$. Substituting into the pass-through formula yields

Result 1 (Pass-through of production and handling costs).

$$\frac{\rho_h}{\rho_c} = r(p), \quad (12)$$

with pass-through into list price

$$\rho_c = \frac{-S'(\mu)}{D'(p) - S'(\mu) \mu_p}, \quad \rho_h = r(p) \rho_c. \quad (13)$$

Both local cost shocks move effective marginal revenue μ , but they do so at different margins. A one-dollar rise in production cost is equivalent to a one-dollar fall in μ on each unit firms supply. A one-dollar rise in handling cost lowers μ by $r(p)$ per gross unit sold, because only returned units incur h .

Market clearing $D(p) = S(\mu)$ translates a given change in μ into list price through the same local comparative statics, regardless of which cost rose. Pass-through therefore scales with the shock's direct impact on μ , yielding $\rho_h/\rho_c = r(p)$. When $r \equiv 0$, handling does not move μ and ρ_c coincides with the textbook competitive pass-through formula.

5 Incidence

Consumer surplus responds to a cost shock only through pass-through,

$$\frac{dCS}{dx} = -D(p)(1 - r(p)) \rho_x. \quad (14)$$

Producer surplus is $PS = \int_0^\mu S(\tilde{\mu}) d\tilde{\mu}$ at equilibrium. Differentiating along the shock gives

$$\frac{dPS}{dx} = \int_0^\mu S_x(\tilde{\mu}, x) d\tilde{\mu} + Q(\mu_p \rho_x + \mu_x), \quad (15)$$

using $PS'(\mu) = Q$. For production cost, $\mu_c = 0$ and $S_c = -S'(\tilde{\mu})$, so

$$\int_0^\mu S_c(\tilde{\mu}) d\tilde{\mu} = -Q, \quad \frac{dPS}{dc} = Q(\mu_p \rho_c - 1). \quad (16)$$

For handling, $S_h = 0$, so

$$\frac{dPS}{dh} = Q(\mu_p \rho_h + \mu_h). \quad (17)$$

The supply-shift term equals the drop in industry quantity at fixed μ , the same $-Q$ that would appear if μ fell by one dollar at fixed p . Both shocks therefore satisfy

$$\frac{dPS}{dx} = Q(\mu_p \rho_x + \mu_x), \quad (18)$$

with $\mu_x = -1$ for production cost and $\mu_x = \mu_h$ for handling.

Write $I_x \equiv |dCS/dx|/|dPS/dx|$. For both shocks,

$$I_c = I_h = \frac{(1 - r(p)) S'(\mu)}{-D'(p)}. \quad (19)$$

6 Tax neutrality

Weyl and Fabinger (2013) show that without returns, buyer and seller remittance deliver the same equilibrium. Fix checkout price p and tax τ . Seller remittance clears at

$$D(p) = S(p - \tau - r(p)(p + h)), \quad (20)$$

because both the return rate and per-return cost $p + h$ are evaluated at checkout price p . Buyer remittance clears at

$$D(p) = S((p - \tau) - r(p - \tau)(p - \tau + h)), \quad (21)$$

because the seller posts list price $p - \tau$, so returns and refund cost are evaluated there. The two conditions define the same equilibrium only if the arguments of $S(\cdot)$ coincide; canceling $p - \tau$ yields

$$r(p)(p + h) = r(p - \tau)(p - \tau + h). \quad (22)$$

Result 2 (Failure of tax neutrality with returns). *Seller and buyer taxes define the same equilibrium if and only if $r(p)(p+h) = r(p-\tau)(p-\tau+h)$.¹ When $r \equiv 0$, both clearings reduce to $D(p) = S(p-\tau)$ and legal incidence is neutral in the Weyl and Fabinger sense.*

Without returns, the tax is only a wedge of τ between payment and producer receipt. With returns, each gross sale still loses τ to the tax, but expected refund cost per sale, $r(p)(p+h)$, depends on the seller's posted list price. When the seller remits the tax, that posted price is checkout price p ; when the buyer remits it, the posted price is $p - \tau$. Legal incidence therefore shifts both $r(\cdot)$ and the refund cost the seller bears on returns, so buyer and seller remittance need not satisfy the same market-clearing condition unless r takes a restrictive form.

¹Write $g(x) \equiv r(x)(x+h)$, so neutrality requires $g(p) = g(p-\tau)$. If $g(x) \equiv C$, then $r(x) = C/(x+h)$; without that functional form on r , the two clearing conditions need not define the same equilibrium.